

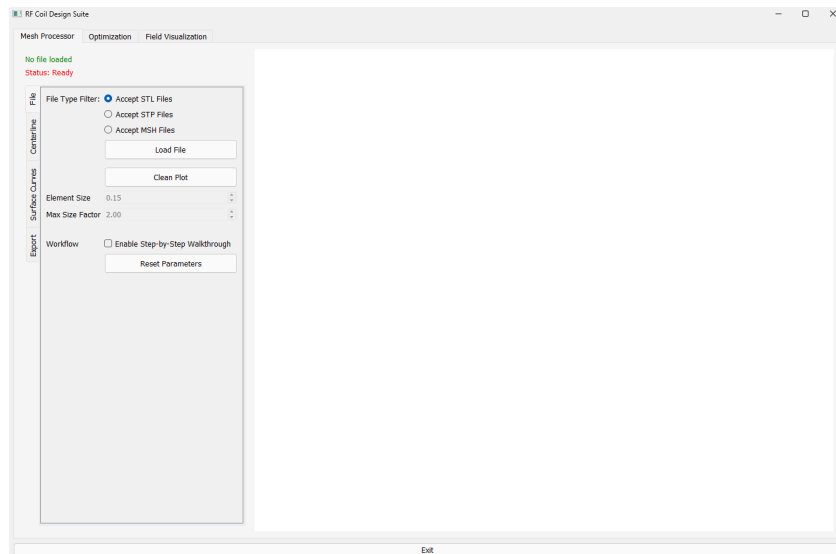
Tutorial of Mesh to Visualization (2026/02/15):

This is a walkthrough of the RF-Coil_B1Field_Visualizer GUI created to observe and analyze the magnetic fields of a coil. Within this document, we will move step by step through a .stp coil; from how to manipulate the meshing to create surface curves, to visualizing the magnetic field of the coil.

Mesh Processor:

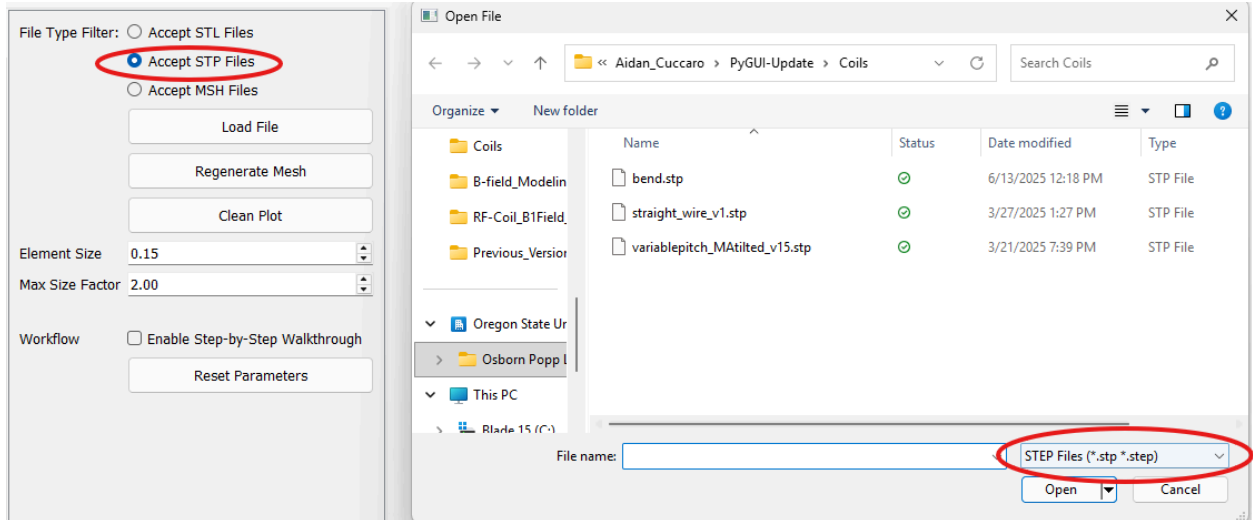
On Startup:

When you first start up the GUI, you will load into the file tab of the mesh processor. This tab starts the process, allowing you to choose the type of file to load, cleaning the current plot, and meshing parameters.



File Filter:

Choose what file can be loaded. This can be an .stl, .stp, or .msh file. The *Load File* button will only display files of the type selected. In our example, we will be using a simple bend.stp file which can be found within the repo as “*example_stp.stp*”. An .stl and .msh version are available as well.

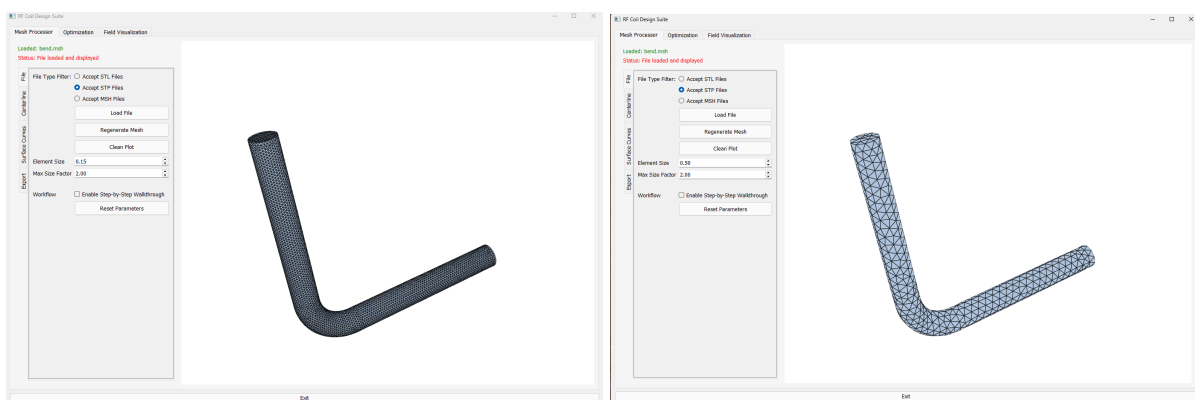


Clean Plot:

Cleans the plot. The current input file is still loaded, but it is removed from the display.

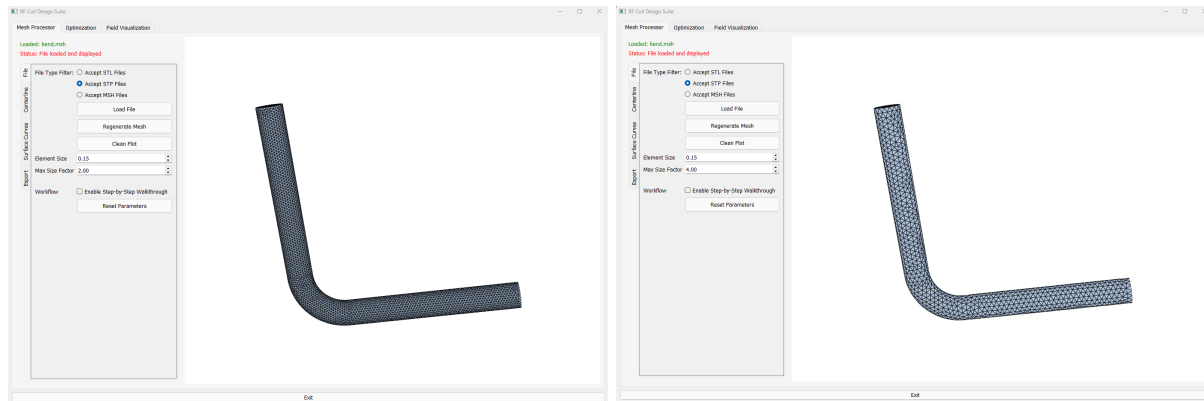
Mesh Parameters:

These parameters can only be changed if a .stp file is loaded. Element size will change the size of vertices and edges, creating a lower poly mesh:



Element Size Comparison of 0.15 to 0.50 Respectively

Max Size Factor increases and decreases the size of the current mesh, in the base case, the size of the triangles:



Max Size Factor of 2.00 to 4.00 Respectively

Workflow:

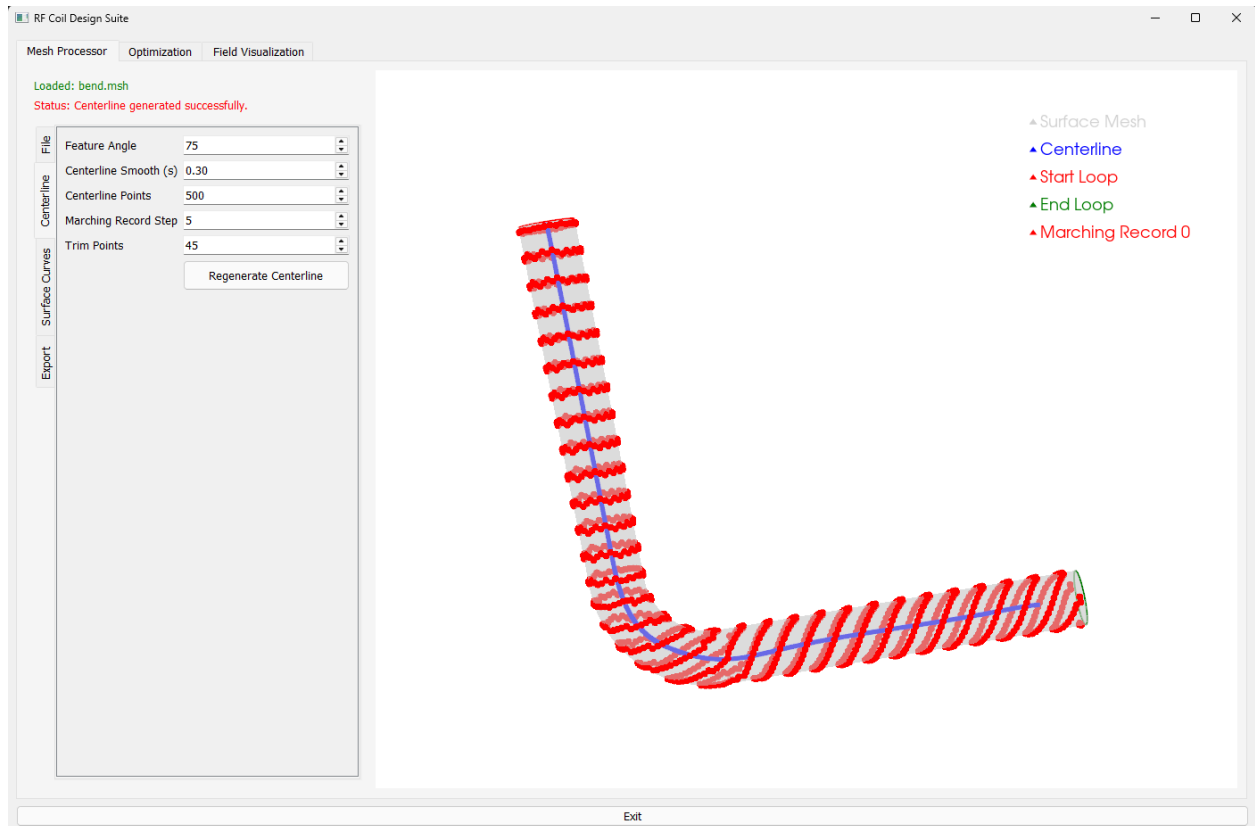
If this is your first time running through the program, the workflow button walks you through the surface curve process, moving you from tab to tab.

Reset Parameters:

The reset parameters button will revert all parameters to their initial default numbers.

Centerline Tab:

Once a mesh has been created, open the centerline tab (enabling walkthrough will take you here immediately). The parameters here will help you generate a centerline that best fits the loaded coil.



Centerline Parameters:

Coils can vary drastically, and as such require different parameter setups. This is probably where most time within this GUI will be spent, as ensuring your centerline is good will be the basis of your surface curves. The first generation of a centerline can take upwards of 5 minutes, but data is cached and sequential runs are much faster.

Feature angle:

The program determines the beginning and end of the coil by finding sharp edges and dictating them as “end” loops. Lower angles mean less sharp angles, and the program will let you go up to 90 degrees. This is the only parameter with a maximum limit.

Centerline Smooth:

Centerline smoothing is a commonly used parameter that spline fits the line such that jagged edges are less common and points are more aligned.

Centerline Points:

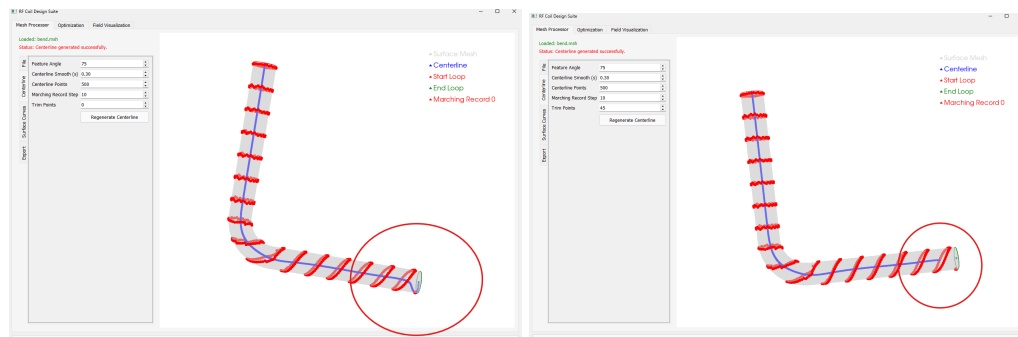
Centerline points dictate how many points will be in the centerline in total. Higher counts means the centerline will be more sampled, will be thicker, and more of a line-shape. Lower samples will have less data points, meaning connections can look more rigid, but may remove impurities that can occur.

Marching Record Step:

The red rings that follow the outside of the coil are the marching rings, which is determined by the marching record step. The centerline is determined by this marching record, which is why you may see “waves” in the centerline as it goes around corners.

Trim Points:

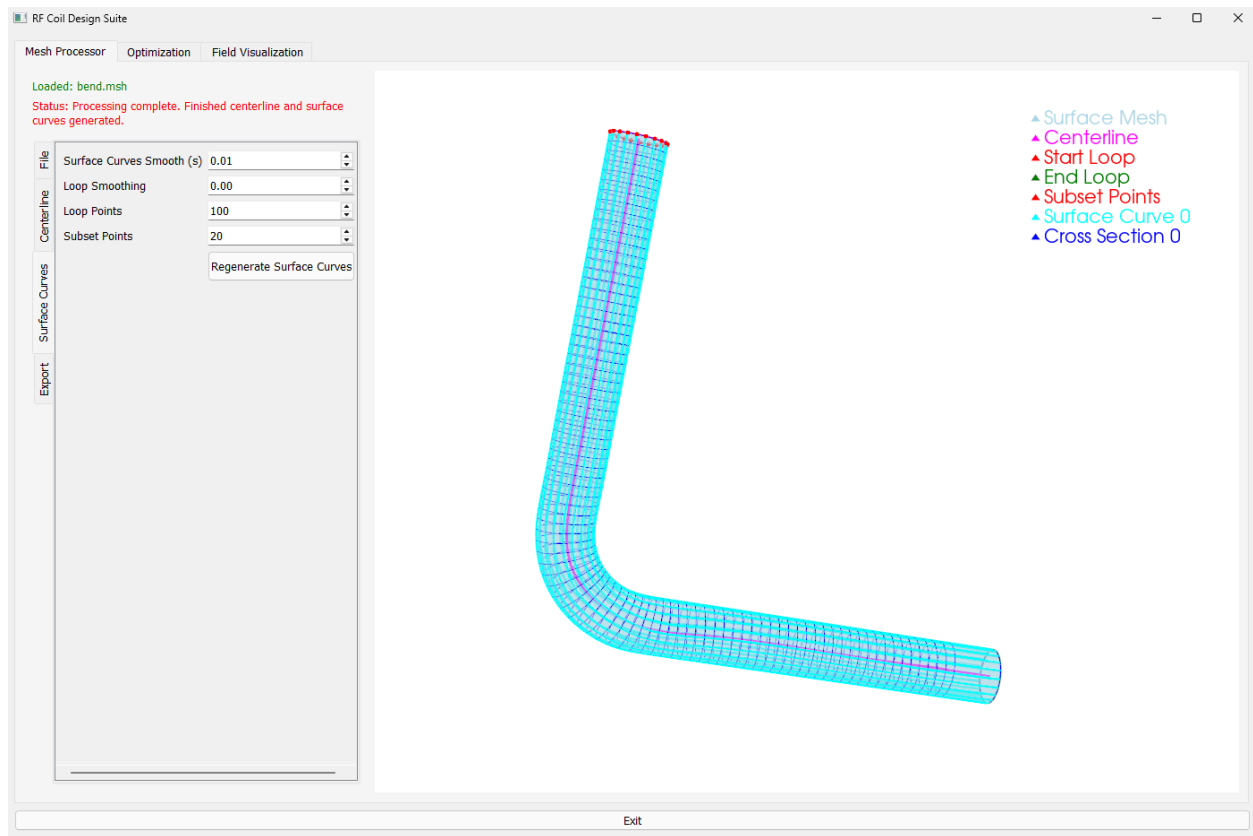
Sometimes, the end of coils shoot off, leading the surface curves to create crazy lines all over the place. Usually, this occurs at the end of the coil because of an inconsistent marching record.



A centerline before and after trimming (left and right respectively)

Surface Curves Tab:

After generating a centerline, the surface curves tab will become available. This final step allows for generation of the actual surface curves along the surface of the coil.



Surface Curves Parameters:

Rarely will these parameters require adjusting, as most of the actual important adjustments will occur before this. These parameters, while minor, can still improve the looks of a coil.

Surface Curves Smooth:

Based on the same algorithm that smooths the centerline, surface curves smoothing will fit surface curves using spline fitting.

Loop Smoothing:

Smooth loops based on the same concept as the other smoothing functions.

Loop Points:

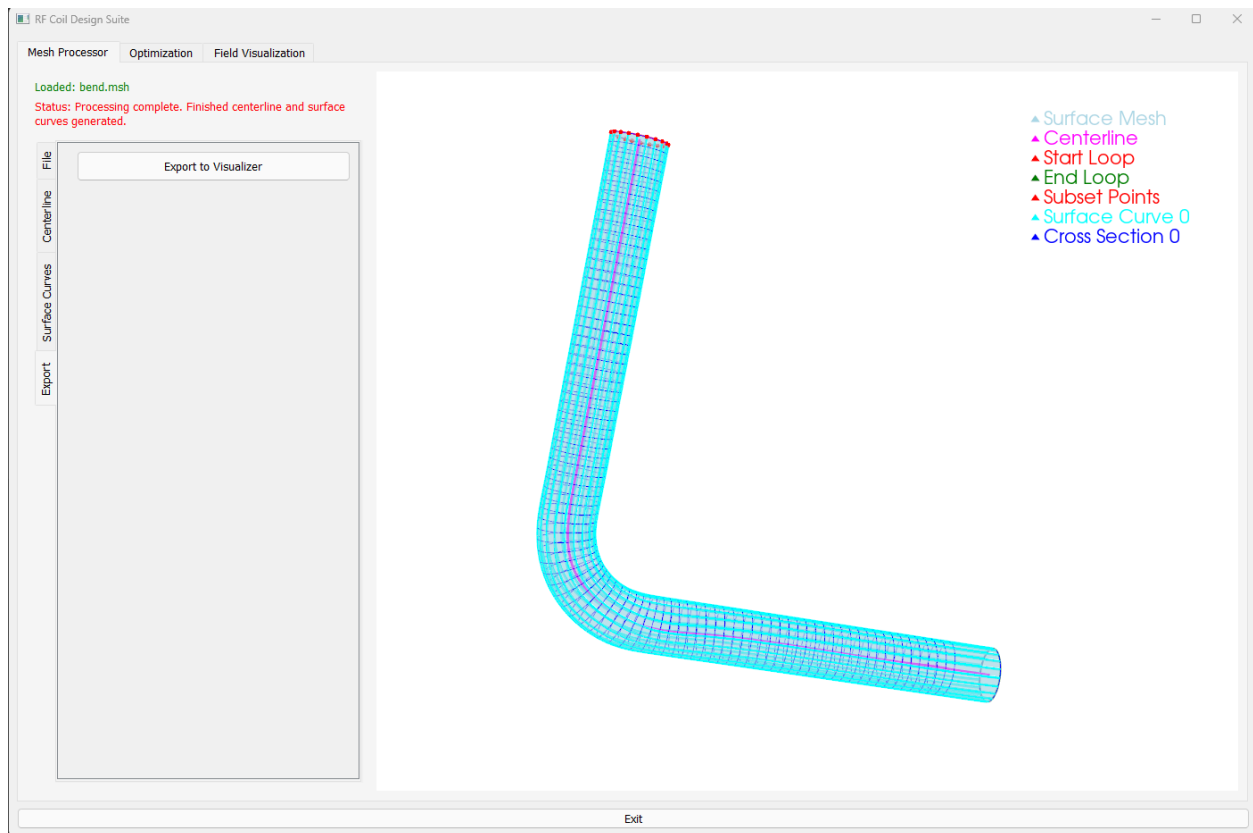
Determines how many points are within the loops.

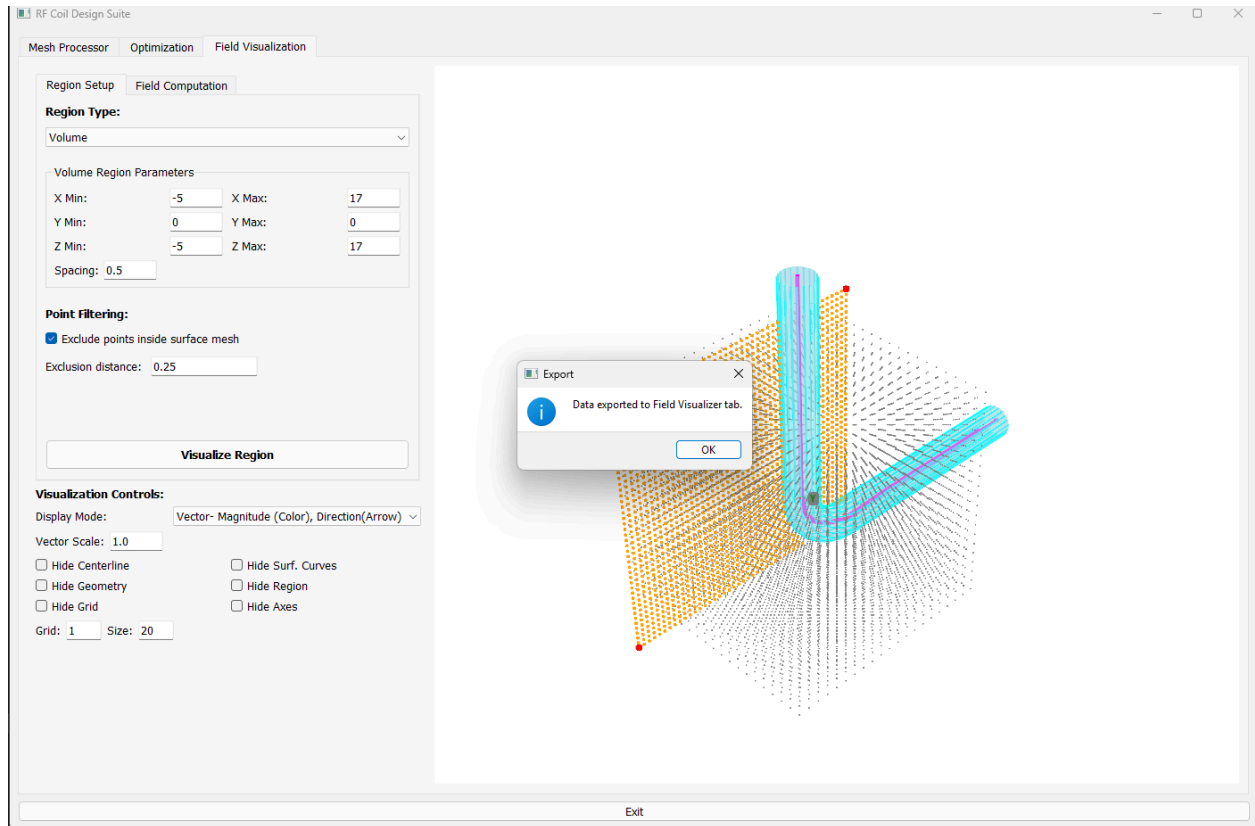
Subset Points:

Determines the number of lines that make up the traces that follow the surface curves across the coil.

Exporting:

Once you are satisfied with the surface curves, move it to the visualizer:

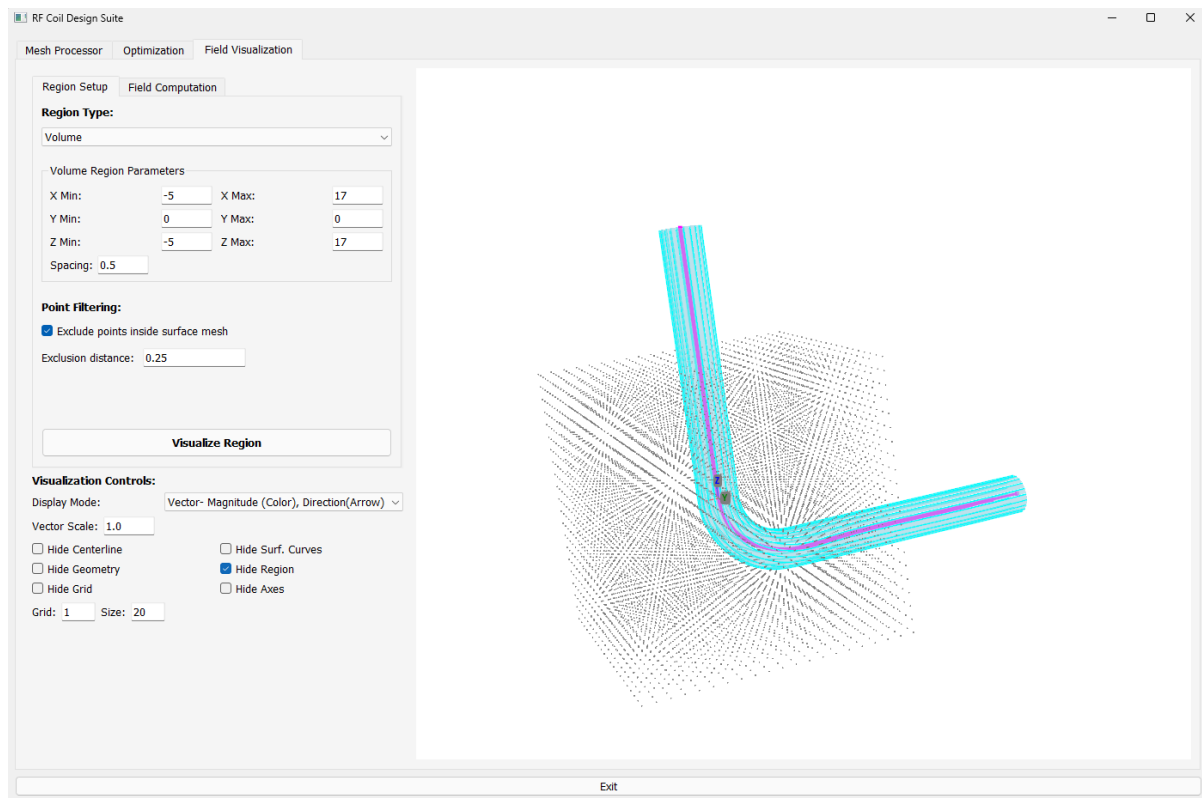
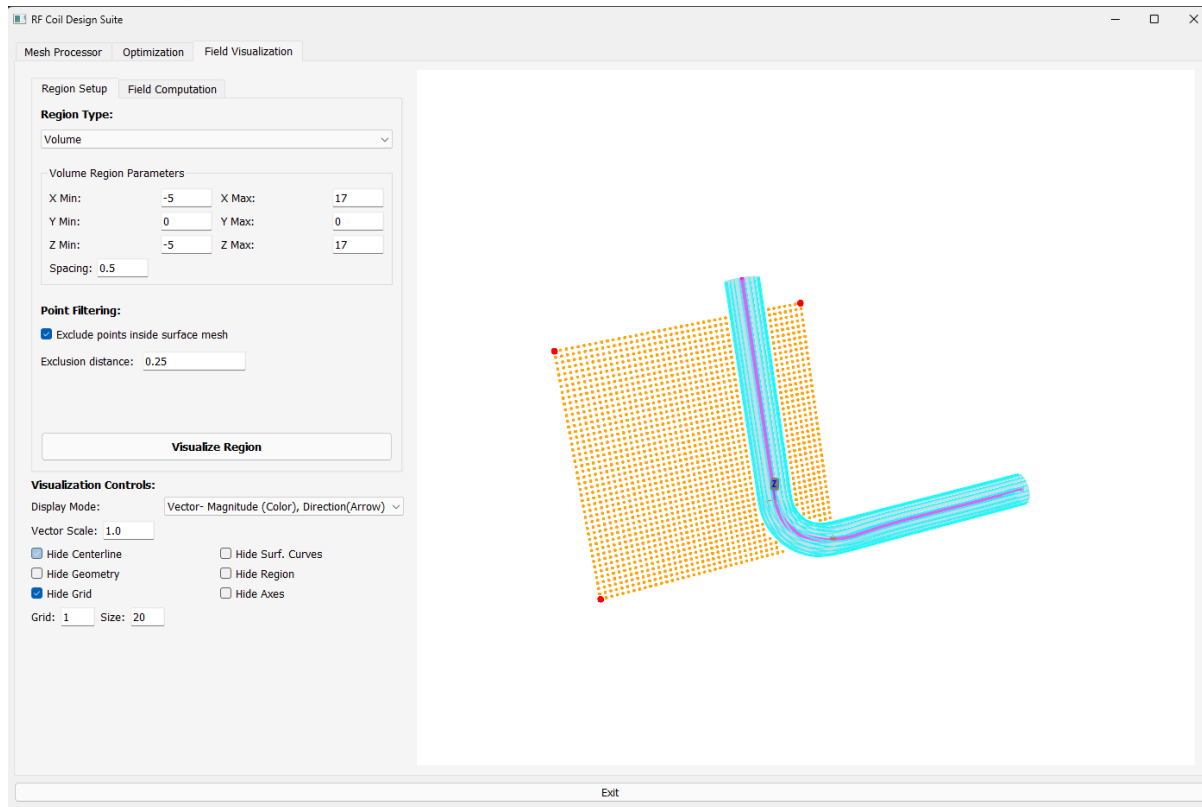




Field Visualization:

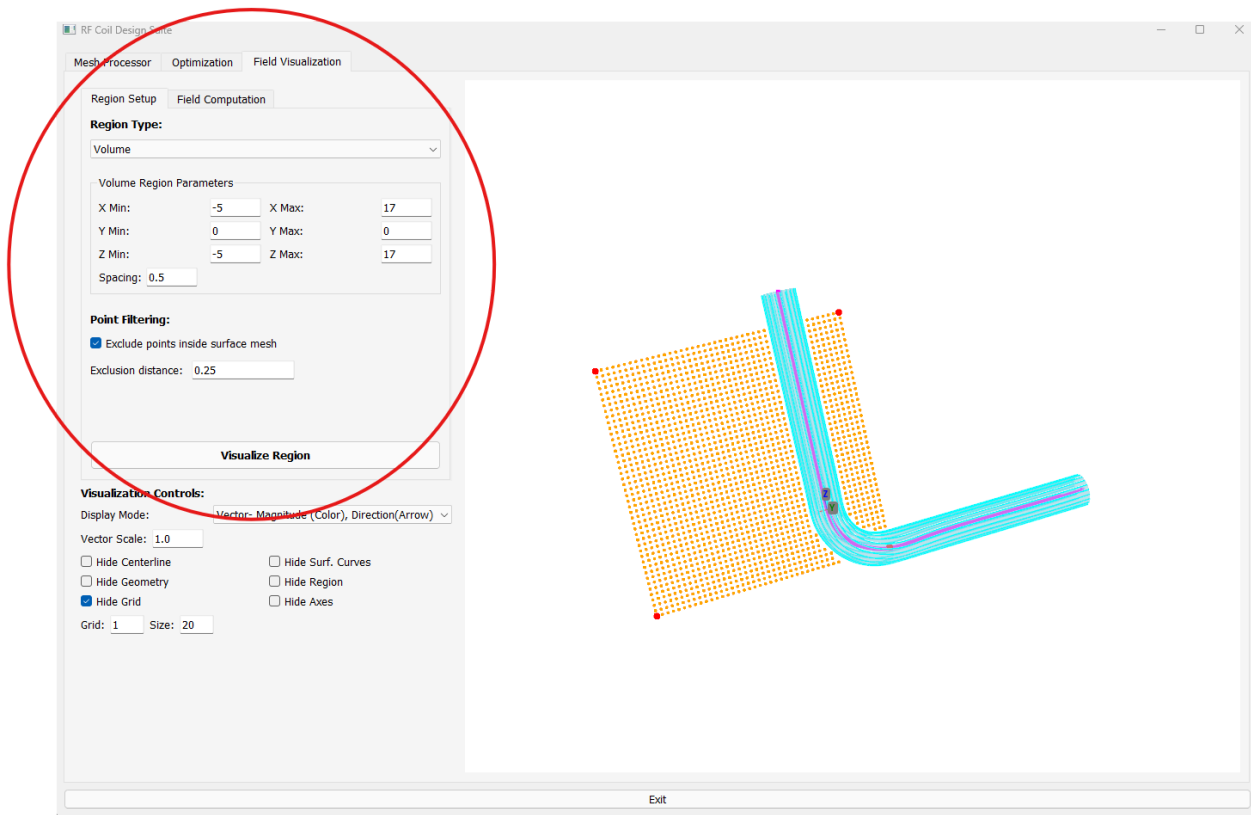
Region Setup:

Once a coil has been created, we can visualize the magnetic field within the field visualization tab. Initially, the program will open the region setup, which manages the grid and the region. Additionally we are able to manipulate the type of vector that will be displayed.

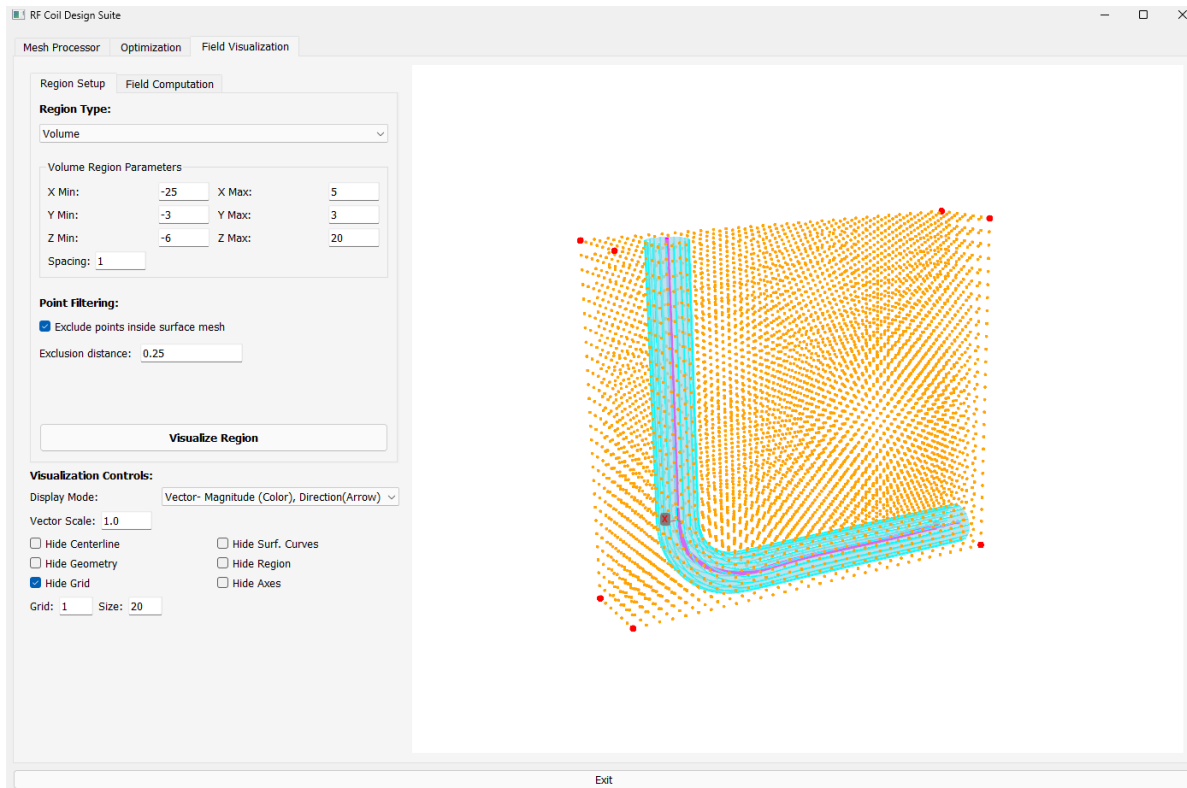


The orange dots are the region while the black dots symbolize the grid.

The region can be manipulated by the Region Controls in the upper right panel:

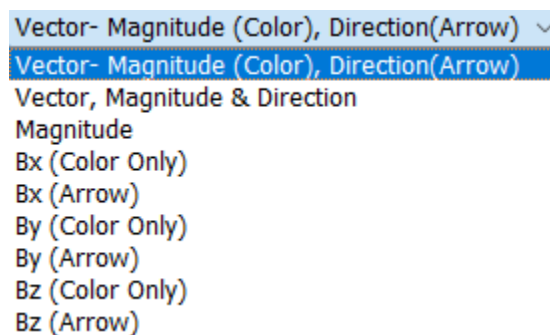


The region can be managed by manipulating the volume coordinates, allowing you to translate the volume to better fit your coil. You can additionally change this region using a plane or line formula as well. The region corresponds to where vectors will stem from when generated, so ensuring that the orange box fits nicely over your coil is important.



A region that has been translated to fit the coil and spacing increased to 1.

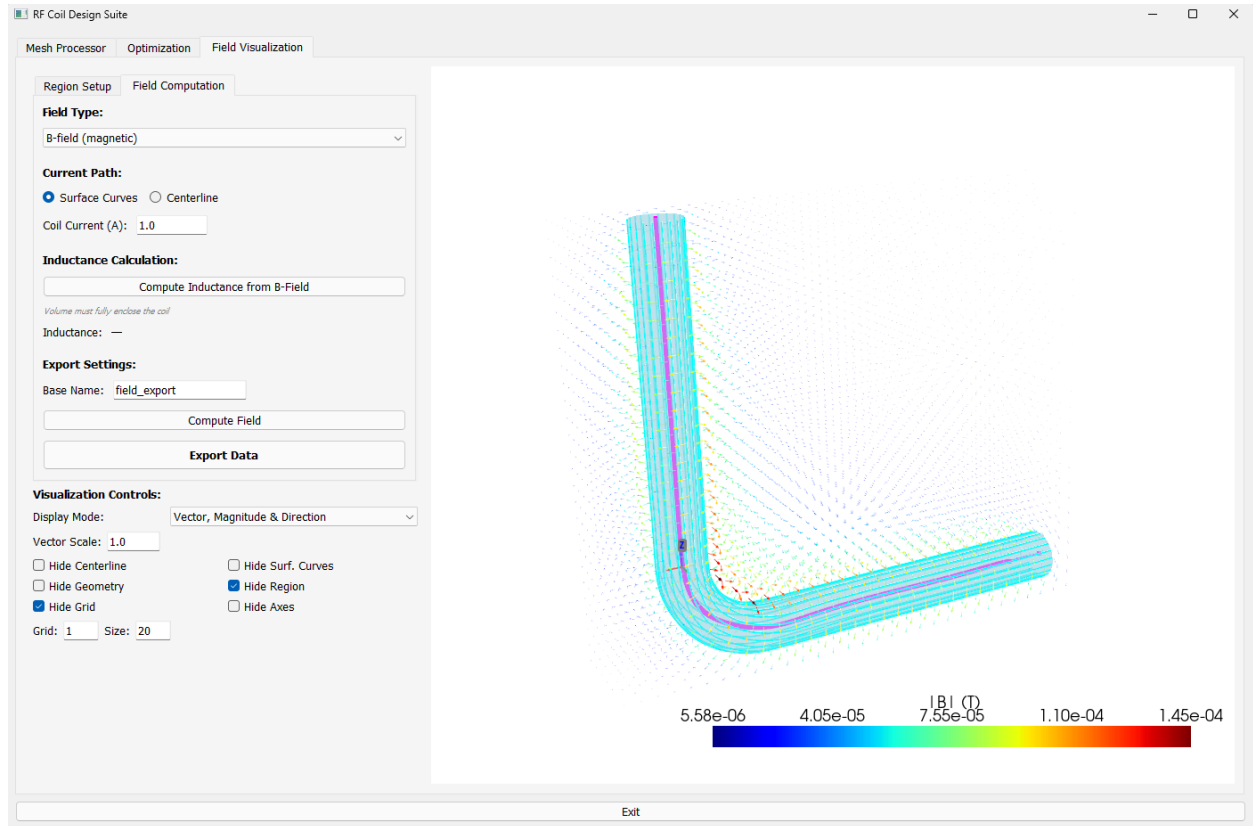
The display mode is also an important feature, as this allows to change the type of vector that is displayed:



By default, Vector- Magnitude (Color), Direction (Arrow) is chosen. The second option, Vector, Magnitude, and Direction, is what will be used for this overview.

Field Computation:

After the region has been moved to the correct location, we can move to field computation. A simple field would look something like the example below:



An example of a magnetic field on a bend coil .stp. The region has been translated over the coil (shown in previous images) and the region and grid are hidden. We have chosen the vector option; Vector, Magnitude, and Direction; which changes the colors of the vectors as well as size based on intensity.

The field computation currently only has B-field (magnetic), but a variety of options, such as following the path of the surface curves or centerline, and inductance calculation can be done. To create the vector field, click “compute field” and the program will display the chosen format. Visualization controls are still available on this tab to allow for hiding various objects such as region (orange dots), grid, axes, and the coil itself.

Once a magnetic field visualization has been completed, the *Export Data* button can be clicked to store the results in a file.

When finished with the GUI, click the *Exit* button at the bottom of the GUI.